Join

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Cartesian (cross join)

	mission_pct	manager_id	department_id	department_id	department_name
1	[NULL]	[NULL]	90	10	Administration
2	[NULL]	100	90	10	Administration
3	[NULL]	100	90	10	Administration
4	[NULL]	102	60	10	Administration
5	[NULL]	103	60	10	Administration

```
select *
        from employees, departments
```

Cartesian with condition

	last_name	department_name
1	King	Executive
2	Yang	Executive
3	Garcia	Executive
4	James	IT
5	Miller	IT
6	Williams	IT
7	Jackson	IT
8	Nguyen	IT
9	Gruenberg	Finance

```
select
        e.last_name,
        d.department_name
        from employees e, departments d
where
        e.department_id=d.department_id
```

Inner join

	last_name	department_name
1	King	Executive
2	Yang	Executive
3	Garcia	Executive
4	James	IT
5	Miller	IT
6	Williams	IT
7	Jackson	IT
8	Nguyen	IT
9	Gruenberg	Finance

```
select
        e.last_name,
        d.department_name
from employees e join departments d on e.department_id=d.department_id
```

Right join

	last_name	department_name
101	Martinez	Marketing
102	Davis	Marketing
103	Jacobs	Human Resources
104	Brown	Public Relations
105	Higgins	Accounting
106	Gietz	Accounting
107	[NULL]	Operations
108	[NULL]	Corporate Tax
109	[NULL]	Benefits

```
select
        e.last_name,
        d.department_name
from employees e right join departments d
                        on e.department_id=d.department_id
```

Left join

	last_name	department_id	department_id	department_name
73	Bates	80	80	Sales
74	Kumar	80	80	Sales
75	Abel	80	80	Sales
76	Hutton	80	80	Sales
77	Taylor	80	80	Sales
78	Livingston	80	80	Sales
79	Grant	[NULL]	[NULL]	[NULL]
80	Johnson	80	80	Sales
81	Taylor	50	50	Shipping
82	Fleaur	50	50	Shipping

```
select
  e.last_name,
  e.department_id,
  d.department_id,
  d.department_name
from employees e left join departments d
                    on e.department_id=d.department_id
```

Full join

	last_name	department_id	department_id	department_name
1	Grant	[NULL]	[NULL]	[NULL]
2	[NULL]	[NULL]	210	IT Support
3	[NULL]	[NULL]	250	Retail Sales
4	[NULL]	[NULL]	160	Benefits
5	[NULL]	[NULL]	120	Treasury
6	[NULL]	[NULL]	260	Recruiting

```
select
  e.last_name,
  e.department_id,
  d.department_id,
  d.department_name
from employees e full join departments d
                 on e.department_id=d.department_id
order by
  case
  when d.department_name is null then 1
  when e.last_name is null then 2
  else d.department_id end
```

Self join

	123 employee_id	A-Z last_name	123 employee_id	A-Z last_name
1	100	King	101	Yang
2	100	King	102	Garcia
3	102	Garcia	103	James
4	103	James	104	Miller
5	103	James	105	Williams
6	103	James	106	Jackson
7	103	James	107	Nguyen
8	101	Yang	108	Gruenberg

```
select
  e.employee_id,
  e.last_name,
  b.employee_id,
  b.last_name
from employees e join employees b
       on e.employee_id=b.manager_id
```

Anti join

	last_name	department_id	department_id
1	Abel	80	10
2	Abel	80	20
3	Abel	80	30
4	Abel	80	40
5	Abel	80	50
6	Abel	80	60
7	Abel	80	70
8	Abel	80	90
9	Abel	80	100
10	Abel	80	110

```
select
  e.last_name,
  e.department_id,
  d.department_id
from employees e join departments d
       on e.department_id != d.department_id
order by 1,3
```

Union
-
bez powtórzeń

	123 department_id
1	[NULL]
2	80
3	210
4	90
5	50
6	220
7	260

```
select department_id from employees e
union
Select department_id from departments d
```

Union all
-
z powtórzeniami

	123 department_id
1	90
2	90
3	90
4	60
5	60
6	60
7	60
8	60

```
select department_id from employees e
union all
select department_id from departments d
```

Intersect
-
część wspólna

	123 department_id
1	20
2	50
3	30
4	110
5	70
6	100
7	80
8	90
9	60
10	10
11	40

```
select department_id from employees e
intersect
select department_id from departments d
```

Except
-
różnica

	123 department_id
1	210
2	220
3	260
4	150
5	160
6	180

```
select department_id from departments d
except
select department_id from employees e
```

Thank you for your attention